Tu Yuang

IELTS: 6.5

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Education

2021.09 - 2025.06 **Shanghai Jiao Tong University** **Energy and Power Engineering** **B.Eng**
Score:79.25/100 GPA:3.05/4.3 Rank:12/22

2025.08 - now **Nanyang Technological University** **Mechanical Engineering** **M.Sc**

Project Experiences

2025.9 ~ now Enhancing Fine-grained Manipulation of Humanoids Using VLA Model

- ***Master's Thesis***

Use **MobileAloha** to collect multimodal data and **fine-tune** VLA model to improve generalization and stability across complex tasks. Evaluate different policies on real-world tasks to compare data efficiency, transferability, and real-time control, forming a more robust and generalizable control strategy.

2021.12 ~ 2023.8 Robomaster Competition (World Champion) Project Team Leader

- ***National University Robot Competition***

In 2022, I led the research, development, and maintenance of the infantry robot's mechanical structure. In 2023, I was responsible for the quadrotor aerial robot project. My work included **drone mechanical design** in SolidWorks, performing lightweight, aerodynamic, and vibration optimizations based on simulation and testing. I also conducted **visual positioning experiments** with the Intel RealSense T265, implementing sensor-fusion algorithms for localization and debugging **VIO/VINS**-based visual positioning performance.

2024.12 ~ 2025.5 Research on Flight Control of Kilogram Quadrotor Cyclocopter

- ***Undergraduate Capstone Project***

Development of flight control algorithm for quadrotor cyclocopter based on **PX4 platform**. Implemented **PID control algorithm** and sensor fusion **EKF algorithm** for the aircraft. Developed the **Gazebo plugin** to implement the **physical effect model** of the rolling wing and completed the **Gazebo simulation model of the aircraft**. Completed **sitl simulation** and **hitl simulation**, and achieved **hovering and attitude control** of the simulation model. Design the **control system hardware** and **electrical system** of the aircraft, and complete the hardware wiring.

2022.5 ~ 2023.9 Research on Modeling, Communication, and Control of SCARA Robot

- ***Participation in Research Program (PRP)***

Designed a **machine-vision–based screw pre-tightening system** for a SCARA robot. Trained a **YOLOv8** model to detect screws and their types. Developed a robot **end-effector** capable of automatic bit switching and implemented a **force-feedback control algorithm** to determine tightening levels. Performed forward/inverse kinematics analysis and completed robot modeling and simulation in **MATLAB**.

Internship

2024.6 ~ 2024.9 Shanghai Ranjing Technology Co., Ltd. Algorithm Engineer

Trained deep learning models and evaluated human skeleton keypoint recognition performance. Developed **rule-based algorithms** for multi-exercise repetition counting and violation detection. Designed image-processing methods for **standing long jump distance estimation.**

Academic Awards

2022	National University Robot Competition RoboMaster	**National First Prize**
2022	Shanghai Jiao Tong University C-class Scholarship	**Scholarship**
2023	National University Robot Competition RoboMaster	**National First Prize**
2024	Shanghai Jiao Tong University Academic Progress Scholarship	**Scholarship**
2025	China College Students' Mechanical Engineering Competition	**National Third Prize**
2025	**NTU School of MAE Graduate Scholarship (~USD 40,000)**	**Scholarship**